HAD6750 Lecture 9: Individual Health Behaviors

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1 Becker & Murphy (1988)

Can we explain addictive behaviors (whether good or bad addictions) in a simple, utility-maximizing framework?

Key insight: Agents maximize utility over time (dynamics), where past consumption affects u(current consumption).

Agent utility:

$$U(t) = u[y(t), c(t), S(t)] \text{ in each } t$$

- What kind of time are we using here?
- c and y are consumption goods, where c might be good or harmful. What should be true about u_c?
- S represents "consumption capital" (addiction stock) that impacts wages: think income/ability to consume. This has a **Law of motion:**

$$\dot{S}(t) = c(t) - \underbrace{\delta}_{\text{depreciation}} S(t) - \underbrace{h[D(t)]}_{\text{investment in treatment}}$$

Agent's problem:

Agents have *forward-looking* utility:

$$\max_{y(t),c(t),D(t)} U(0) = \int_0^T e^{-\sigma t} u[y(t), c(t), S(t)]dt \text{ subject to } \dot{S}(t), \text{budget constraint,}$$

where the budget constraint is given by:

$$\int_0^T e^{-rt}[\underbrace{1\, y(t) + p_c(t)c(t) + p_D(t)D(t)}_{\text{all expenditures}}]dt \leq \underbrace{A_0}_{\text{initial assets}} + \int_0^T e^{-rt} \underbrace{w(S(t))}_{\text{wages (formation of addiction)}} dt$$

- Why is $S(\cdot)$ not a choice here? Where does $D(t)$ show up?
- How do we find the optimal paths here? [see supplement]

Solving the Model

To solve this one, we write a version of the Lagrangian that acknowledges that one of the constraints is a **differential equation**, called a **Hamiltonian**:

$$\begin{aligned}\mathcal{H} &= e^{-\sigma t}u(y(t), c(t), S(t)) \\ &+ \lambda\left[A_0 + \int_0^T w(S(t))dt - \int_0^T e^{-rt}[y(t) + p_c(t)c(t) + p_D(t)D(t)]dt\right] \\ &+ \mu(t)[c(t) - \delta S(t) - h(D(t))].\end{aligned}$$

Then, we can take the FOCs as usual, but with a bit of care:

1. $\frac{\partial \mathcal{H}}{\partial y(t)} = e^{-\sigma t}\frac{\partial u}{\partial y} - \lambda e^{-rt} = 0$. This equation governs the marginal utility of $y(t)$.
2. $\frac{\partial \mathcal{H}}{\partial c(t)} = e^{-\sigma t}\frac{\partial u}{\partial c} - \lambda e^{-rt}p_c\, c(t) + \mu(t) = 0$. This equation governs the MU of $c(t)$.
3. $\dot{\mu} = -\frac{\partial \mathcal{H}}{\partial S(t)} = -\left[e^{-\sigma t}\frac{\partial u}{\partial S} + \lambda e^{-rt}\frac{\partial w}{\partial S} - \mu(t)\delta\right]$. Note that this is a first order ODE to get optimal "investment in addiction capital".
4. $\frac{\partial \mathcal{H}}{\partial D(t)} = -\lambda e^{-rt}p_D(t) - \mu(t)\frac{\partial h}{\partial D}(t) = 0$. This equation identifies optimal treatment choice.

What would an equilibrium mean in this context? That is, what do we hope from $(c^*(t), y^*(t), D^*(t))$? One option is a **steady state**, where $\dot{S}(t) = 0$.

- Given this, can you precisely define what the utility would be?

Under quadratic utility,

$$S^*(t) = \frac{(\sigma + \delta)\mu p_c}{2(\delta^2 + \sigma\delta + \gamma - \sigma - 2\delta)}$$

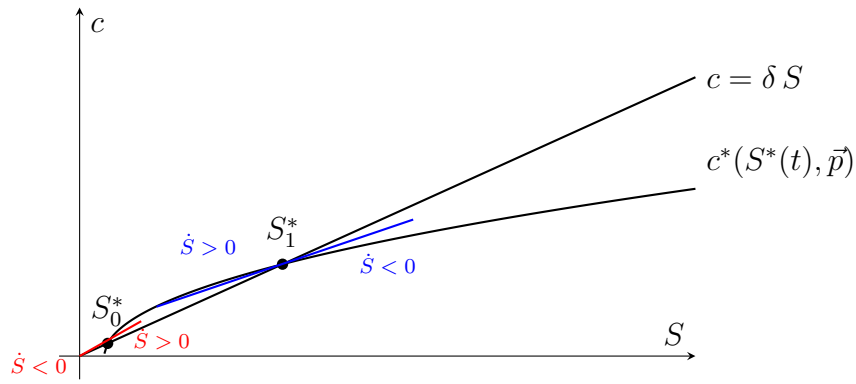
with $\dot{S} = \ddot{S} = 0$. In this world, there are two possible equilibria (given that there are positive and negative roots from setting this equal to 0. Then, S^* and the relative prices pin down $s^*(t), c^*(t)$.

Characterizing the Equilibrium

Throughout, let's assume that $c(\cdot)$ is a bad, addictive good, for illustration. That is, as $c(\cdot)$ increases, so does $S(\cdot)$, and we assume that $U_S < 0$ [how?].

Let's simplify now $\sigma = r$, $T = \infty$, $D(t) = 0$. Also assume u is quadratic. What do each of these mean? What do you think of these assumptions?

Now we can explore the interesting question of **dynamics** and whether or not a given starting point will "arrive" at an equilibrium/steady state. That is, do paths $(y^*(t), c^*(t))$ converge to a steady state, especially for different starting points $y(0), c(0)$?



- What does the straight line indicate? [This is our steady state condition!]
- What does the shift of the $c^*(\cdot)$ curve when $c^*(\cdot) = 0$ mean? [This is the initial stock of addiction capital]
- What is different about these two different equilibria? Which of the steady states are viable or *stable*?
- Imagine indifference curves here—which equilibrium gives higher utility?

Are these Steady States viable (stable)?

1. S_0^*: No. Imagine indifference curves here... want to get to S_1^*. (stable $c \to 0$ w/ time)
2. S_1^*: Yes. This is the addicted equilibrium. Is it bad? What does addiction mean? [If $c \uparrow$ today, then $c \uparrow$ for all future periods].

Comparative Statics

You can consider 2 important types of comparative statics. Can you draw the figures for these yourself?

1. **Change in** δ: what does this mean intuitively? What kind of policy would induce this? What does this imply about where the steady states are, and where they are relative to each other? [Could this eliminate the addicted equilibrium?]
2. **Change in** p_c: c^* shifts to the right. What does this imply for equilibria? Can it capture the concept of **"quitting cold turkey"**?

2 Oster, Shoulson, & Dorsey (2013): Optimal Expectations

This is a really cool model built to explain some unique empirical findings. The key idea is that utility might be **anticipatory**, meaning that too much knowledge about the future could be bad.

Setup

Let utility be defined by the following **payoff matrix**:

$u(a,s)$	$s=0$	$s=1$
$a=0$	1	$-\Omega$
$a=1$	$1-\Phi$	0

Where:

- s: health state (1 = sick, 0 = healthy)
- a: protective action (which depends on beliefs)
- Assume that $\Phi, \Omega < 1$. What does this capture? [Φ is the loss from "protective action"; Ω is the loss from "freewheeling"]. Which would you model as larger?
- Each individual has an exogenous probability of being sick, $p = \mathbb{E}[s]$. This is revealed by testing and unaffected by behavior. [Where is this model useful? Where is it not?]

Model Timing

The model has 3 periods:

0: Testing chosen (cost of c).

1: Beliefs (π) and preventive action a <u>chosen</u>.

2: State s revealed, utility $u(a,s)$ generated (state-dependent utility).

How do you solve something like this? **Backward induction**. There are thus 3 choices that we need to consider:

1. **Choice of a, conditional on π:** Easy! Chosen to maximize $E[u(a,s)|\pi]$.

$$E[u(a,s)|\pi] = \pi\, u(a,1) + (1-\pi)\, u(a,0).$$

   Hence, choose $a=1$ if and only if:

$$\begin{aligned} \pi\, u(1,1) + (1-\pi)\, u(1,0) &\geq \pi\, u(0,1) + (1-\pi)\, u(0,0) \\ \Rightarrow (1-\pi)\,(1-\Phi) &\geq -\pi\,\Omega + (1-\pi) \\ \Rightarrow \pi &\geq \frac{\Omega}{\Phi+\Omega} \end{aligned}$$

   What does this mean? [Beliefs have to be above a certain amount to justify action.]

2. **How are beliefs π chosen in the absence of testing?** Here, we model utility as depending on an (exogenous?) anticipation value:

$$U(\pi|p) = \delta E[u(\hat{a},s)|\pi] + E[u(\hat{a},s)|p].$$

[Note: $\hat{a}(\pi) \neq \hat{a}(p)$. What does this capture?] π depends on ex-post utility and (discounted) subjective utility.

The optimal belief here is going to depend on what action we want to induce. However, we can say in all cases that if you get to choose your belief (you aren't tested), then $\pi \leq p$ always, and in almost all cases this inequality is strict. How do we prove this?

Proof: First, notice that

$$U = \begin{cases} \delta(1-\pi) + (1-p) - (\delta\pi + p)\Omega & \hat{a}(\pi) = 0 \\ [\delta(1-\pi) + (1-p)](1-\Phi) & \hat{a}(\pi) = 1 \end{cases}$$

If we want $\hat{a} = 0$ (from the above step), then we simply let $\hat{\pi} = 0$. If, on the other hand, we want $\hat{a} = 1$, then we can set beliefs directly at $\hat{\pi} = \frac{\Omega}{\Phi+\Omega}$.

Given the utility, it is optimal to do the former ($\hat{a} = 0$) if and only if

$$\begin{aligned} \delta + (1-p) - p\Omega &\geq \left(\delta\left(\frac{\Omega}{\Phi+\Omega}\right) + (1-p)\right)(1-\Phi) \\ \Rightarrow \overline{p} &\leq \frac{\Omega}{\Phi+\Omega} + \frac{\delta\Phi(1+\Omega)}{(\Omega+\Phi)^2}. \end{aligned}$$

Hence, we decide based on the exogenous probability of p. If p is low enough (below $\overline{p}$, then we can set $\hat{a} = 0, \hat{\pi} = 0$ (which is always $< p$). If the individual is high enough risk, then choose the baseline beliefs just high enough to induce protective action.

3. **How is testing chosen?** Testing eliminates choice of π, forcing $\pi = p$. Define the value of testing as:

$$V_{\text{test}} = \begin{cases} p(\Omega - \delta) - c & p \leq \overline{p} \\ \frac{\delta\Phi(1+\Omega)}{\Phi+\Omega} - p(\Omega+\delta) + \Phi - c & p > \overline{p} \end{cases}$$

Whether an individual gets tested is going to depend on, once again, the relative **anticipation value**, here compared against Ω [what's the story?]:

- **Case 1**: If $\delta < \Omega$ [what does this mean?] then testing may have some positive value:

$$V > 0 \Leftrightarrow \begin{cases} p(\Omega - \delta) > c & p \leq \overline{p} \\ \delta\frac{\Phi(1+\Omega)}{\Phi+\Omega} - p(\delta+\Omega) > c & p > \overline{p} \end{cases}$$

  That is, if anticipation value is low enough, then no matter what the exogenous probability of disease is, there is a condition where the cost of testing is low enough to induce takeup of testing.

- **Case 2**: If $\delta > \Omega$, then testing has **negative value** that is **decreasing in** p. This essentially says that when utility places enough weight on the future, exogenous risk *lowers* the likelihood of future screening and that beliefs have power.

Illustration & Interpretation

Here, where is testing optimal? The figure picks some values for losses, probabilities, and δ calibrated to data.

Panel A: **Time 1 anticipatory utility.** Consider difference between the anticipatory utility *with* testing ($\delta(1-p)$), and anticipatory utility *without* testing, which is δ when $p < \overline{p}$ and $0.5\,\delta(1-\Phi)$ otherwise. Up to $\overline{p}$, utility without testing is constant (since people are just acting as if they are healthy and experiencing anticipation associated with that state), and utility with testing is **decreasing** in p. The difference, $-\delta\, p$, is therefore also decreasing in p.

Panel B: **Time 2 consumption utility.** The difference between realized utility with testing $(1-p)$ and realized utility without testing $(-p\Phi + (1-p)$ if $p < \overline{p}$ and $(1-p)(1-\Phi)$ otherwise). Utility both with and without testing is **decreasing** in p, but the utility without testing is decreasing <u>faster</u>.

Panel C: **Putting them together.** Value of testing is increasing in risk, but with incorporating anticipatory utility, this value is much lower than it would be based **only** on consumption utility. Even with small c, testing can be discouraged just because of the loss of the anticipatory utility

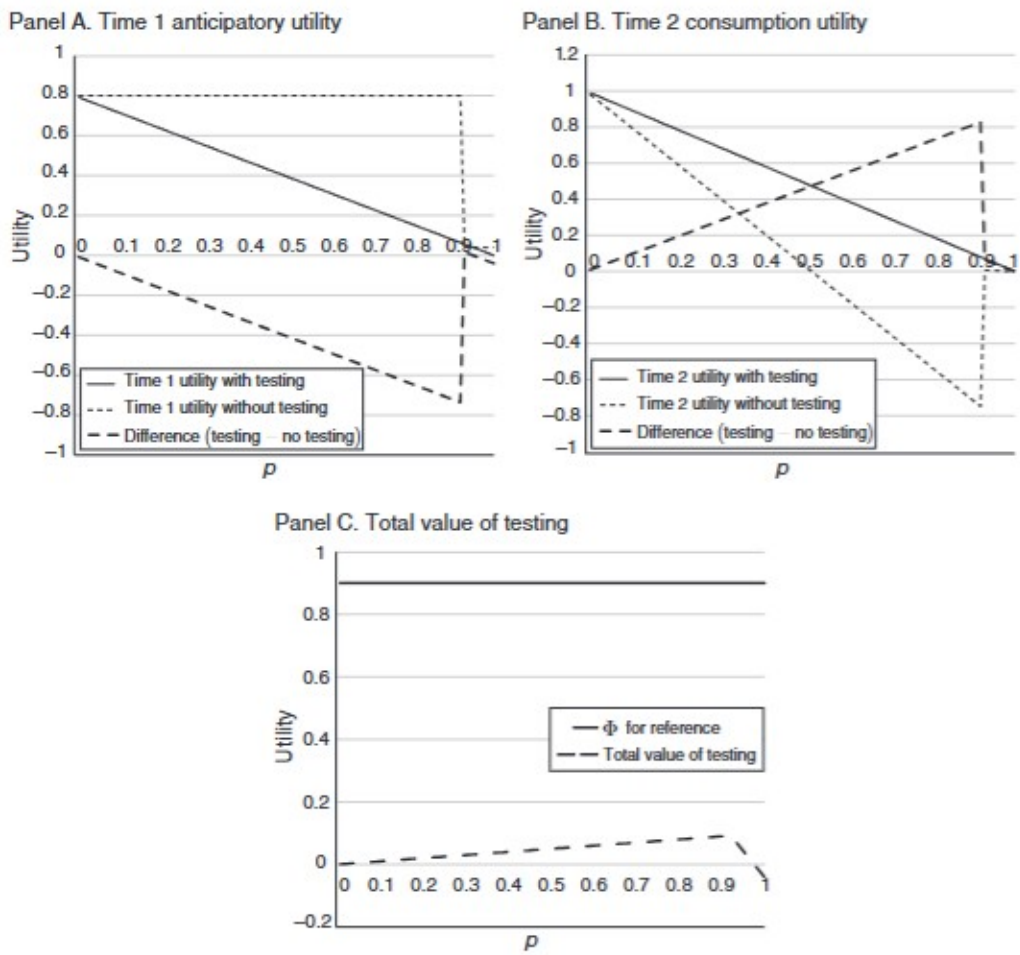


FIGURE 6. OPTIMAL EXPECTATIONS MODEL, GRAPHICAL INTUITION

Notes: These figures give intuition for the optimal expectation result on testing. Panel A shows the utility in the anticipatory period with and without testing. Panel B shows utility in the consumption period, with and without testing. In both cases the thicker dotted line shows the difference between the two utilities. Panel C shows the sum of the utilities; this figure also includes (for reference) a line showing the value of $\Phi = 0.9$. These graphs are made assuming symmetric losses from the wrong action ($\Phi = \Omega$) and assuming that $\delta < \Phi$. Under the chosen values of δ and Φ we have $p^* = 0.92$.

How do we think about the value of this model? It is very specific, no?

3 Cutler & Donahoe

[NOTE: Underdeveloped Section]

Baseline: rational addiction model, where utility is expanded to:

$$V(a_{it}, S_{it}, x_{it}, c_{it}, \mathbb{E}[\bar{a}_t])$$

Where:

- a_{it}: consumption of addictive goods (c_{it} is now normal consumption good)
- x_{it}: taste parameters

- $\mathbb{E}[\overline{a}_t]$: peer consumption.

In addition, we have stock evolving according to:

$$S_{i,t+1} = (1-d)(S_{it} + a_{it}),$$

so that there is a complementarity between addiction stock and addictive consumption.

Given the same setup as above and quadratic u, we get demand:

$$a^*_{it} = \alpha\, S_{it} + \gamma\, \bar{a}_t + \pi\, p_t + \delta\, x_{it} + k$$

- α: addictiveness parameter
- $\gamma\, \bar{a}_t$: captures **spillover effects** of addiction ($\gamma = \frac{b_g}{b_{aa}}$)
- Supply shocks are captured by p_t and demand shocks by x_{it}

You can extend this model to include death with some hazard function

$$H(a_{it}, S_{it}), \text{ with } H_a > 0, H_s > 0, H_{as} < 0.$$

- If $\gamma = 0$, then epidemics continue (so there is no steady state) if and only if individuals are sufficiently addicted:
$$\frac{b_{as}(1-d)}{d} > b_{aa}.$$
- If $\gamma \neq 0$, so that there are spillovers, this becomes **easier** to satisfy (meaning it is harder to stop epidemics):
$$\frac{b_{as}(1-d)}{d} + b_g > b_{aa}.$$